

Previous Year Questions

QUES

2 Mark Questions

Q1: Express 156 as a product of its prime factors. [PYQ] (2 Marks)

- The prime factorization is performed by dividing the number by the smallest prime numbers sequentially.
- $156 = 2 \times 78$
- $78 = 2 \times 39$
- $39 = 3 \times 13$
- Therefore, $156 = 2^2 \times 3 \times 13$.

Q2: Given that $HCF(306, 657) = 9$, find $LCM(306, 657)$. [PYQ] (2 Marks)

- We use the property: $HCF(a, b) \times LCM(a, b) = a \times b$.
- Substitute the given values: $9 \times LCM = 306 \times 657$.
- $LCM = \frac{306 \times 657}{9} = 34 \times 657$.
- The LCM is 22,338.

Q3: Check whether 6^n can end with the digit 0 for any natural number n . (2 Marks)

- For a number to end with the digit 0, its prime factorization must contain both 2 and 5 as factors.
- The prime factorization of 6^n is $(2 \times 3)^n = 2^n \times 3^n$.
- Since 5 is not a prime factor of 6^n , the number cannot end with the digit 0 for any value of n .

Q11: Find the HCF of 96 and 404 by the prime factorization method. [PYQ] (2 Marks)

- Prime factorization of $96 = 2^5 \times 3$.
- Prime factorization of $404 = 2^2 \times 101$.

- HCF is the product of the lowest powers of common factors: $2^2 = 4$.
- Therefore, the HCF is 4.

Q12: Find the LCM and HCF of 6 and 20 by the prime factorization method. (2 Marks)

- Factorize the numbers: $6 = 2 \times 3$ and $20 = 2^2 \times 5$.
- HCF (lowest power of common factor) $= 2^1 = 2$.
- LCM (highest power of all factors) $= 2^2 \times 3 \times 5 = 60$.
- The HCF is 2 and the LCM is 60.

Q13: Why does the number 4^n , where n is a natural number, never end with the digit 0? (2 Marks)

- A number ends in 0 only if its prime factors include both 2 and 5.
- The prime factorization of 4^n is $(2^2)^n = 2^{2n}$.
- Since 5 is not a factor in the factorization, 4^n can never end with the digit 0.

Q14: Explain why $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is a composite number. (2 Marks)

- Factor out 5 from the expression: $5(7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1)$.
- The expression simplifies to $5(1008 + 1) = 5 \times 1009$.
- Since it is a product of two numbers other than 1 and itself, it is a composite number.

Q15: If $HCF(26, 169) = 13$, find $LCM(26, 169)$. (2 Marks)

- Relationship: $LCM = (a \times b)/HCF$.
- Calculation: $LCM = (26 \times 169)/13$.
- Result: $LCM = 2 \times 169 = 338$.
- The LCM is 338.

Q16: Find the HCF of 510 and 92. [PYQ] (2 Marks)

- $510 = 2 \times 3 \times 5 \times 17$.
- $92 = 2^2 \times 23$.
- Common factor is 2; the lowest power is 2^1 .

- HCF is 2.

Q17: Check if 12^n can end with the digit 0 or 5 for any natural number n . (2 Marks)

- Prime factorization of $12^n = (2^2 \times 3)^n = 2^{2n} \times 3^n$.
- To end in 0, it needs factors 2 and 5. To end in 5, it needs factor 5 but not 2.
- Since 5 is not a prime factor of 12^n , it can never end with 0 or 5.

Q18: Find the HCF and LCM of 17, 23, and 29. (2 Marks)

- The numbers 17, 23, and 29 are all prime numbers.
- HCF of any set of prime numbers is always 1.
- LCM is the product of the numbers: $17 \times 23 \times 29 = 11339$.

Q19: Find the HCF of 72 and 120. [PYQ] (2 Marks)

- $72 = 2^3 \times 3^2$.
- $120 = 2^3 \times 3 \times 5$.
- $\text{HCF} = 2^3 \times 3 = 8 \times 3 = 24$.
- The HCF is 24.



Q20: Write the prime factorization of 3825. [PYQ] (2 Marks)

- Divide by 3: $3825/3 = 1275$.
- Divide by 3: $1275/3 = 425$.
- Divide by 5: $425/5 = 85$. Divide by 5: $85/5 = 17$.
- The prime factorization is $3^2 \times 5^2 \times 17$.

3 Mark Questions

Q4: Prove that $\sqrt{3}$ is an irrational number. [PYQ] (3 Marks)

- Assumption: Let $\sqrt{3}$ be a rational number of the form a/b , where a and b are co-prime integers and $b \neq 0$.
- Squaring both sides: $3 = a^2/b^2 \implies a^2 = 3b^2$. This means 3 divides a^2 , and by theorem, 3 divides a .

- Substitution: Let $a = 3c$. Then $(3c)^2 = 3b^2 \implies 9c^2 = 3b^2 \implies b^2 = 3c^2$.
- Conclusion: This implies 3 divides b^2 , and thus 3 divides b .
- Contradiction: Since a and b both have 3 as a common factor, they are not co-prime. Our initial assumption is wrong; therefore, $\sqrt{3}$ is irrational.

Q5: Prove that $3 + 2\sqrt{5}$ is irrational. [PYQ] (3 Marks)

- Let $3 + 2\sqrt{5}$ be rational, such that $3 + 2\sqrt{5} = a/b$, where a, b are integers and $b \neq 0$.
- Rearranging the terms: $2\sqrt{5} = (a/b) - 3 \implies 2\sqrt{5} = (a - 3b)/b$.
- Isolating the radical: $\sqrt{5} = (a - 3b)/2b$.
- Logical Deduction: Since a and b are integers, $(a - 3b)/2b$ is a rational number.
- Contradiction: This implies $\sqrt{5}$ is rational, which contradicts the known fact that $\sqrt{5}$ is irrational. Thus, $3 + 2\sqrt{5}$ is irrational.

Q6: Find the HCF and LCM of 12, 15, and 21 using the prime factorization method. (3 Marks)

- Prime factorization: $12 = 2^2 \times 3$, $15 = 3 \times 5$, and $21 = 3 \times 7$.
- HCF (Highest Common Factor): Identify the lowest power of common factors. Common factor is 3. $HCF = 3^1 = 3$.
- LCM (Least Common Multiple): Identify the highest power of all prime factors involved. $LCM = 2^2 \times 3 \times 5 \times 7$.
- Calculation: $LCM = 4 \times 3 \times 5 \times 7 = 420$.

Q21: Prove that $2 + \sqrt{3}$ is an irrational number. [PYQ] (3 Marks)

- Let $2 + \sqrt{3}$ be rational, so $2 + \sqrt{3} = a/b$ where a, b are integers.
- Rearrange: $\sqrt{3} = (a/b) - 2 = (a - 2b)/b$.
- Since a and b are integers, $(a - 2b)/b$ is a rational number.
- This implies $\sqrt{3}$ is rational, which contradicts the fact that $\sqrt{3}$ is irrational.
- Thus, $2 + \sqrt{3}$ is irrational.

Q22: Find the LCM and HCF of 8, 9, and 25 by prime factorization. (3 Marks)

- $8 = 2^3$, $9 = 3^2$, $25 = 5^2$.
- HCF: There are no common prime factors, so $HCF = 1$.
- LCM: $2^3 \times 3^2 \times 5^2 = 8 \times 9 \times 25 = 1800$.
- The HCF is 1 and LCM is 1800.

Q23: There is a circular path around a sports field. Sonia takes 18 minutes to drive one round, while Ravi takes 12 minutes. If they start at the same point, after how many minutes will they meet again at the starting point? [PYQ] (3 Marks)

- The time they meet again is the LCM of 18 and 12.
- $18 = 2 \times 3^2$ and $12 = 2^2 \times 3$.
- $LCM = 2^2 \times 3^2 = 4 \times 9 = 36$.
- They will meet again after 36 minutes.

Q24: Prove that $\sqrt{2}$ is irrational. (3 Marks)

- Assume $\sqrt{2} = a/b$ (co-prime). $2 = a^2/b^2 \implies a^2 = 2b^2$.
- 2 divides a^2 , so 2 divides a . Let $a = 2c$.
- $(2c)^2 = 2b^2 \implies 4c^2 = 2b^2 \implies b^2 = 2c^2$.
- 2 divides b^2 , so 2 divides b . Since a and b share factor 2, they aren't co-prime.
- Therefore, $\sqrt{2}$ is irrational.

Q25: Show that $5\sqrt{2}$ is irrational. [PYQ] (3 Marks)

- Let $5\sqrt{2} = a/b$ (rational).
- $\sqrt{2} = a/5b$.
- Since a and b are integers, $a/5b$ is rational.
- But $\sqrt{2}$ is known to be irrational.
- This contradiction proves $5\sqrt{2}$ is irrational.

Q26: The HCF of two numbers is 23 and their LCM is 1449. If one number is 161, find the other number. (3 Marks)

- Use the formula: $OtherNumber = (HCF \times LCM)/FirstNumber$.

- Calculation: $(23 \times 1449)/161$.
- Simplify: $1449/7 = 207$.
- The other number is 207.

Q27: Prove that $7 - \sqrt{5}$ is irrational. (3 Marks)

- Let $7 - \sqrt{5} = r$ (rational).
- $\sqrt{5} = 7 - r$.
- Rational subtracted from rational is rational, so $7 - r$ is rational.
- But $\sqrt{5}$ is irrational. This is a contradiction.
- Hence, $7 - \sqrt{5}$ is irrational.

Q28: Find the smallest number which when divided by 28 and 32 leaves remainders 8 and 12 respectively. (3 Marks)

- Difference: $28 - 8 = 20$ and $32 - 12 = 20$. The common difference is 20.
- Required number = $LCM(28, 32) - 20$.
- $28 = 2^2 \times 7$ and $32 = 2^5$. $LCM = 2^5 \times 7 = 224$.
- Number = $224 - 20 = 204$.

Q29: Show that $3\sqrt{7}$ is irrational. [PYQ] (3 Marks)

- Let $3\sqrt{7} = a/b$ (rational).
- $\sqrt{7} = a/3b$.
- Since a, b are integers, $a/3b$ is rational.
- But $\sqrt{7}$ is irrational, hence a contradiction.
- Therefore, $3\sqrt{7}$ is irrational.

Q30: Find the HCF and LCM of 144 and 198 using the Fundamental Theorem of Arithmetic. (3 Marks)

- $144 = 2^4 \times 3^2$.
- $198 = 2 \times 3^2 \times 11$.
- $HCF = 2 \times 3^2 = 18$.

- $LCM = 2^4 \times 3^2 \times 11 = 16 \times 9 \times 11 = 1584$.

5 Mark Questions

Q7: (a) Prove that $\sqrt{5}$ is irrational. (b) Hence, show that $3 - 2\sqrt{5}$ is irrational. [PYQ] (5 Marks)

- Part (a): Assume $\sqrt{5} = a/b$ (co-prime). Squaring gives $5b^2 = a^2$. Thus 5 divides a . Let $a = 5c$, then $5b^2 = 25c^2 \implies b^2 = 5c^2$, so 5 divides b . This contradicts the co-prime assumption.
- Part (b): Assume $3 - 2\sqrt{5}$ is rational, say x . Then $x = 3 - 2\sqrt{5}$.
- Rearrange the equation to isolate the irrational part: $2\sqrt{5} = 3 - x$.
- Divide by 2: $\sqrt{5} = (3 - x)/2$.
- Conclusion: Since x is rational, $(3 - x)/2$ must be rational. This contradicts the result from Part (a). Therefore, $3 - 2\sqrt{5}$ must be irrational.

Q8: State the Fundamental Theorem of Arithmetic and use it to find the HCF and LCM of 336 and 54. Verify that $HCF \times LCM = Product$ of the numbers. [PYQ] (5 Marks)

- Definition: The Fundamental Theorem of Arithmetic states that every composite number can be expressed as a product of primes, and this factorization is unique apart from the order of factors.
- Factorization: $336 = 2^4 \times 3 \times 7$ and $54 = 2 \times 3^3$.
- HCF and LCM: $HCF = 2^1 \times 3^1 = 6$. $LCM = 2^4 \times 3^3 \times 7 = 16 \times 27 \times 7 = 3024$.
- Verification Step 1: $HCF \times LCM = 6 \times 3024 = 18,144$.
- Verification Step 2: Product of numbers = $336 \times 54 = 18,144$. Since LHS = RHS, the relationship is verified.

Q9: A sweetseller has 420 kaju barfis and 130 badam barfis. She wants to stack them such that each stack has the same number and takes up the least area of the tray. How many barfis can be placed in each stack? (5 Marks)

- Problem Analysis: To minimize the area used, the number of barfis in each stack must be the maximum possible that divides both 420 and 130.

- Mathematical Goal: We need to find the Highest Common Factor (HCF) of 420 and 130.
- Prime Factorization: $420 = 2^2 \times 3 \times 5 \times 7$ and $130 = 2 \times 5 \times 13$.
- Finding HCF: The common prime factors are 2 and 5. The lowest power of 2 is 2^1 and 5 is 5^1 .
- Result: $HCF = 2 \times 5 = 10$. The sweetseller can make stacks of 10 barfis each.

Q10: Find the largest number which divides 70 and 125, leaving remainders 5 and 8, respectively. (5 Marks)

- Step 1: Subtract the remainders from the given numbers. $70 - 5 = 65$ and $125 - 8 = 117$.
- Step 2: The required number is the HCF of 65 and 117.
- Step 3: Prime factorization: $65 = 5 \times 13$ and $117 = 3^2 \times 13$.
- Step 4: Identify common factors: The only common prime factor is 13.
- Final Answer: The HCF is 13. Therefore, 13 is the largest number that divides 70 and 125 leaving remainders 5 and 8.

Q31: State the Fundamental Theorem of Arithmetic. Using this, find the HCF and LCM of 4052 and 12576. [PYQ] (5 Marks)

- Definition: Every composite number can be uniquely factorized as a product of primes, ignoring the order.
- Factorization: $4052 = 2^2 \times 1013$; $12576 = 2^5 \times 3 \times 131$.
- HCF: Smallest power of common factor (2) is $2^2 = 4$.
- LCM: $2^5 \times 3 \times 131 \times 1013 = 12739392$.
- Conclusion: HCF is 4 and LCM is 12,739,392.

Q32: Prove that for any prime number p , $\sqrt{p}$ is irrational. (5 Marks)

- Assumption: Let $\sqrt{p} = a/b$ where a, b are co-primes.
- Step 1: $p = a^2/b^2 \implies a^2 = pb^2$. This means p divides a^2 , so p divides a .
- Step 2: Let $a = pk$. Then $(pk)^2 = pb^2 \implies p^2k^2 = pb^2 \implies b^2 = pk^2$.
- Step 3: This means p divides b^2 , so p divides b .

- Conflict: a and b share common factor p , contradicting co-prime status.
- Final: Therefore, $\sqrt{p}$ is irrational.

Q33: Can two numbers have 18 as their HCF and 380 as their LCM? Give reasons. (5 Marks)

- Rule: The HCF must always be a factor of the LCM.
- Division: $380/18$.
- Calculation: $380 = 18 \times 21 + 2$.
- Observation: 18 does not divide 380 exactly (remainder is 2).
- Conclusion: No, two numbers cannot have 18 as HCF and 380 as LCM because the HCF does not divide the LCM.

Q34: An army contingent of 616 members is to march behind an army band of 32 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march? [PYQ] (5 Marks)

- Concept: To find the maximum number of columns, we must find the $HCF(616, 32)$.
- Factorization 616: $2^3 \times 7 \times 11$.
- Factorization 32: 2^5 .
- HCF: The common factor is 2; the lowest power is $2^3 = 8$.
- Answer: The maximum number of columns they can march in is 8.

Q35: Prove that the sum or difference of a rational and an irrational number is irrational. Use $5 + \sqrt{2}$ as an example. (5 Marks)

- Let $5 + \sqrt{2}$ be rational, say x .
- Isolate the radical: $\sqrt{2} = x - 5$.
- Logic: x is rational, and 5 is rational. The difference of two rational numbers is always rational.
- Contradiction: This implies $\sqrt{2}$ is rational, but we know $\sqrt{2}$ is irrational.
- Conclusion: Our assumption that $5 + \sqrt{2}$ is rational is false. Thus, it is irrational.

Q36: Find the HCF and LCM of 120, 144, and 160 using prime factorization method. [PYQ] (5 Marks)

- $120 = 2^3 \times 3 \times 5$.
- $144 = 2^4 \times 3^2$.
- $160 = 2^5 \times 5$.
- HCF: Common factor is 2; lowest power is $2^3 = 8$.
- LCM: Highest powers are $2^5, 3^2, 5$. $LCM = 32 \times 9 \times 5 = 1440$.

Q37: Show that the square of any positive integer is of the form $3m$ or $3m + 1$ for some integer m . [PYQ] (5 Marks)

- Let a be any positive integer. By division lemma, $a = 3q, 3q + 1$, or $3q + 2$.
- Case 1: $a = 3q \implies a^2 = 9q^2 = 3(3q^2) = 3m$.
- Case 2: $a = 3q + 1 \implies a^2 = (3q + 1)^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1 = 3m + 1$.
- Case 3: $a = 3q + 2 \implies a^2 = (3q + 2)^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1 = 3m + 1$.
- Conclusion: Therefore, the square of any positive integer is of the form $3m$ or $3m + 1$.

Q38: (a) Find HCF(96, 404) by prime factorization. (b) Hence find their LCM. (c) Verify $HCF \times LCM = Product$. [PYQ] (5 Marks)

- Part (a): $96 = 2^5 \times 3, 404 = 2^2 \times 101$. $HCF = 2^2 = 4$.
- Part (b): $LCM = (96 \times 404)/4 = 24 \times 404 = 9696$.
- Part (c): $HCF \times LCM = 4 \times 9696 = 38784$.
- Product: $96 \times 404 = 38784$.
- Result: Verified.

Q39: Prove that $1/\sqrt{3}$ is irrational. If $\sqrt{3} = 1.732$, find the value of $1/\sqrt{3}$ up to 3 decimal places. (5 Marks)

- Let $1/\sqrt{3} = a/b$ (rational). $\sqrt{3} = b/a$. Since b/a is rational, $\sqrt{3}$ is rational, which is a contradiction. Hence $1/\sqrt{3}$ is irrational.
- Calculation: $1/1.732 = 1000/1732$.

- Simplify: $1000/1732 \approx 0.5773\dots$
- Answer: 0.577.

Q40: Find the HCF of 65 and 117 and express it in the form $65m + 117n$. (5 Marks)

- HCF calculation: $65 = 5 \times 13$ and $117 = 3^2 \times 13$. $HCF = 13$.
- Using Euclid's Division Lemma: $117 = 1 \times 65 + 52$ and $65 = 1 \times 52 + 13$.
- Expressing HCF: $13 = 65 - 1 \times 52$.
- Substitute 52: $13 = 65 - (117 - 1 \times 65)$.
- Result: $13 = 2(65) - 1(117)$. Here $m = 2, n = -1$.

Q41: Show that the square of an odd positive integer is of the form $8m + 1$, for some integer m . (5 Marks)

- Any odd positive integer is of the form $4q + 1$ or $4q + 3$.
- Case 1: $(4q + 1)^2 = 16q^2 + 8q + 1 = 8(2q^2 + q) + 1 = 8m + 1$.
- Case 2: $(4q + 3)^2 = 16q^2 + 24q + 9 = 16q^2 + 24q + 8 + 1 = 8(2q^2 + 3q + 1) + 1 = 8m + 1$.
- Conclusion: In both cases, the square is of the form $8m + 1$.

Q42: Three bells toll at intervals of 9, 12, 15 minutes. If they start tolling together, after what time will they next toll together? (5 Marks)

- Concept: The next time they toll together is the $LCM(9, 12, 15)$.
- Factorization: $9 = 3^2, 12 = 2^2 \times 3, 15 = 3 \times 5$.
- LCM: $2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$.
- Conversion: 180 minutes = 3 hours.
- Result: They will next toll together after 3 hours.

Q43: Prove that $2\sqrt{3} + 5$ is irrational, given that $\sqrt{3}$ is irrational. (5 Marks)

- Assume $2\sqrt{3} + 5 = r$ (rational).
- Subtract 5: $2\sqrt{3} = r - 5$.
- Divide by 2: $\sqrt{3} = (r - 5)/2$.
- Logic: Since r is rational, $(r - 5)/2$ is rational.

- Contradiction: But $\sqrt{3}$ is irrational. Hence $2\sqrt{3} + 5$ is irrational.

Q44: Find the largest number that will divide 398, 436 and 542 leaving remainders 7, 11 and 15 respectively. (5 Marks)

- Subtract remainders: $398 - 7 = 391$, $436 - 11 = 425$, $542 - 15 = 527$.
- Find HCF(391, 425, 527).
- $391 = 17 \times 23$; $425 = 5^2 \times 17$; $527 = 17 \times 31$.
- HCF: The common factor is 17.
- Answer: The largest number is 17.

Q45: Show that the cube of any positive integer is of the form $9m$, $9m + 1$ or $9m + 8$. [PYQ] (5 Marks)

- Let $a = 3q + r$ where $r = 0, 1, 2$.
- Case 1 ($r = 0$): $a^3 = (3q)^3 = 27q^3 = 9(3q^3) = 9m$.
- Case 2 ($r = 1$): $a^3 = (3q + 1)^3 = 27q^3 + 27q^2 + 9q + 1 = 9(3q^3 + 3q^2 + q) + 1 = 9m + 1$.
- Case 3 ($r = 2$): $a^3 = (3q + 2)^3 = 27q^3 + 54q^2 + 36q + 8 = 9(3q^3 + 6q^2 + 4q) + 8 = 9m + 8$.
- Final: The cube is always of the form $9m$, $9m + 1$ or $9m + 8$.